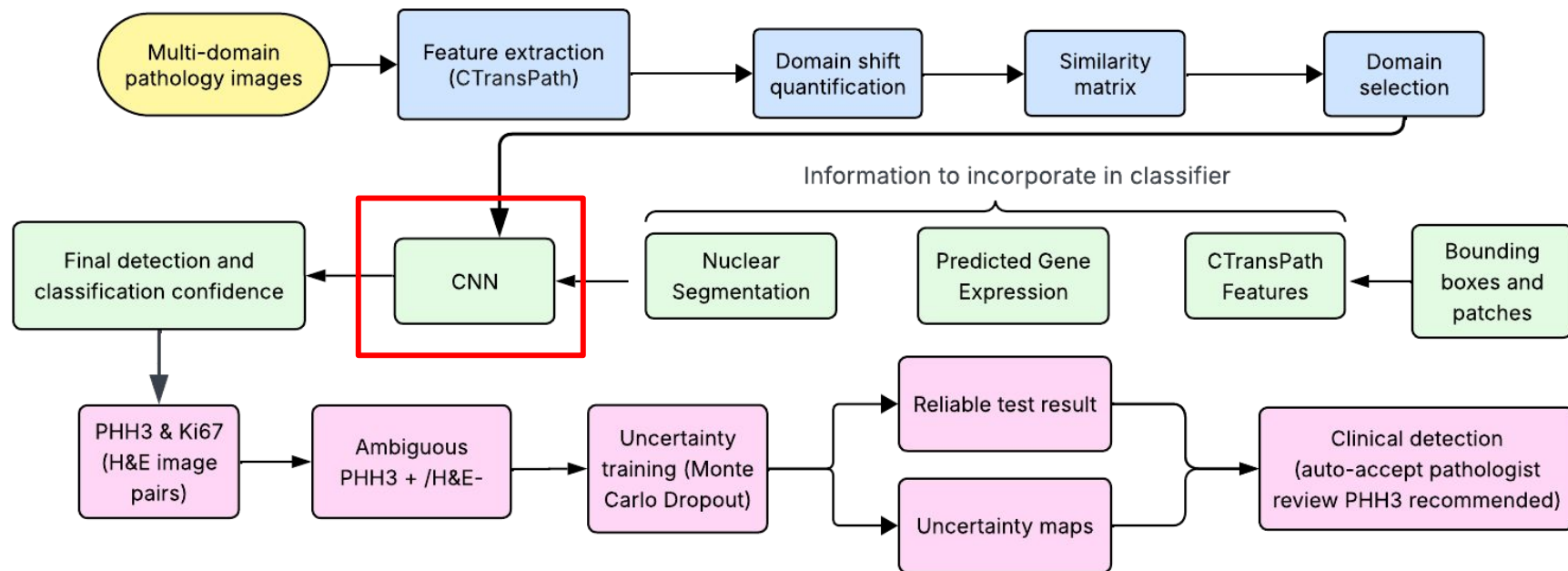




DANN Update

Naina Misra | Lillian Lam | Yang Cao | 3/31/2026

Pipeline



Recap (Adversarial Penalty)

Feature Extractor: learns image representations

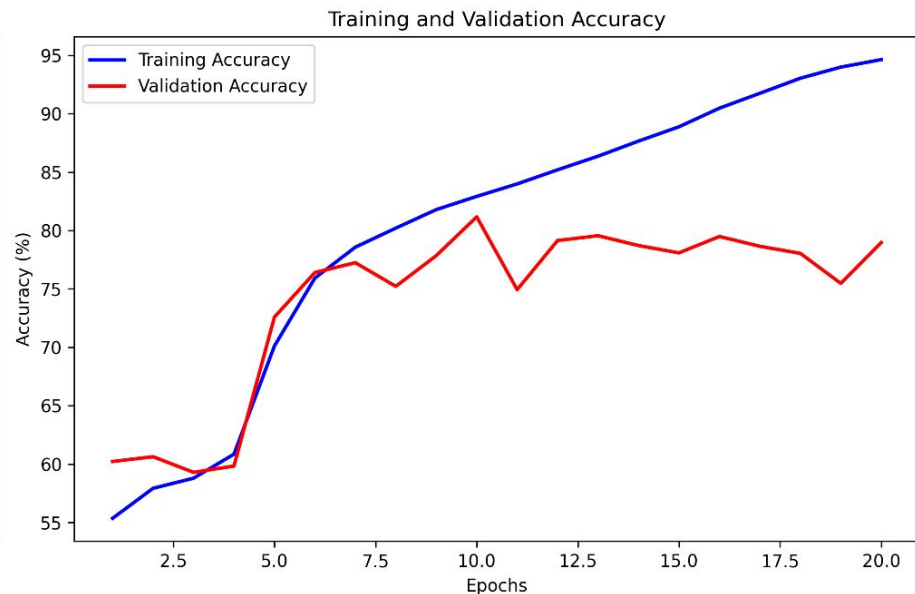
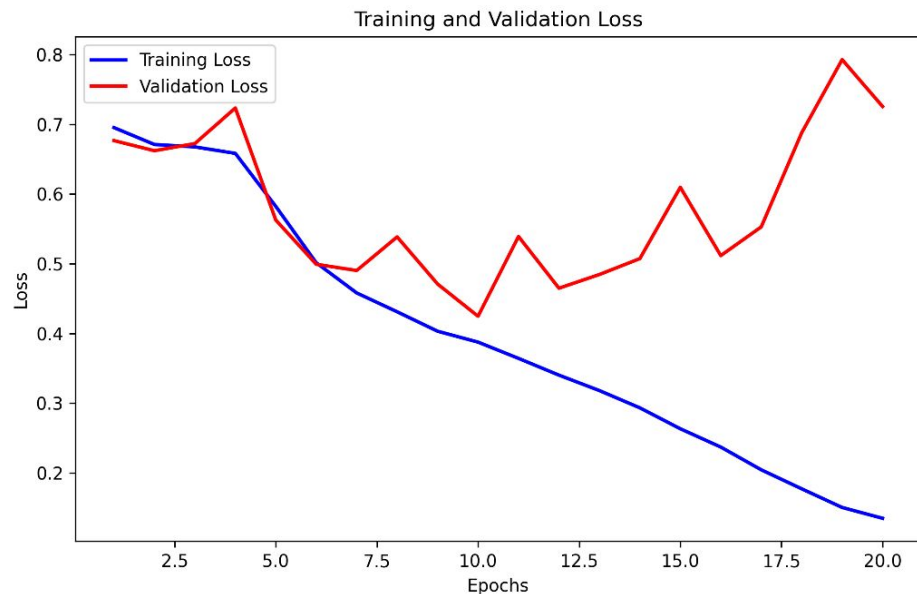
Mitosis Classifier: predicts mitotic vs non-mitotic (primary task)

Domain Classifier: tries to predict which slide the patch came from (adversarial task)

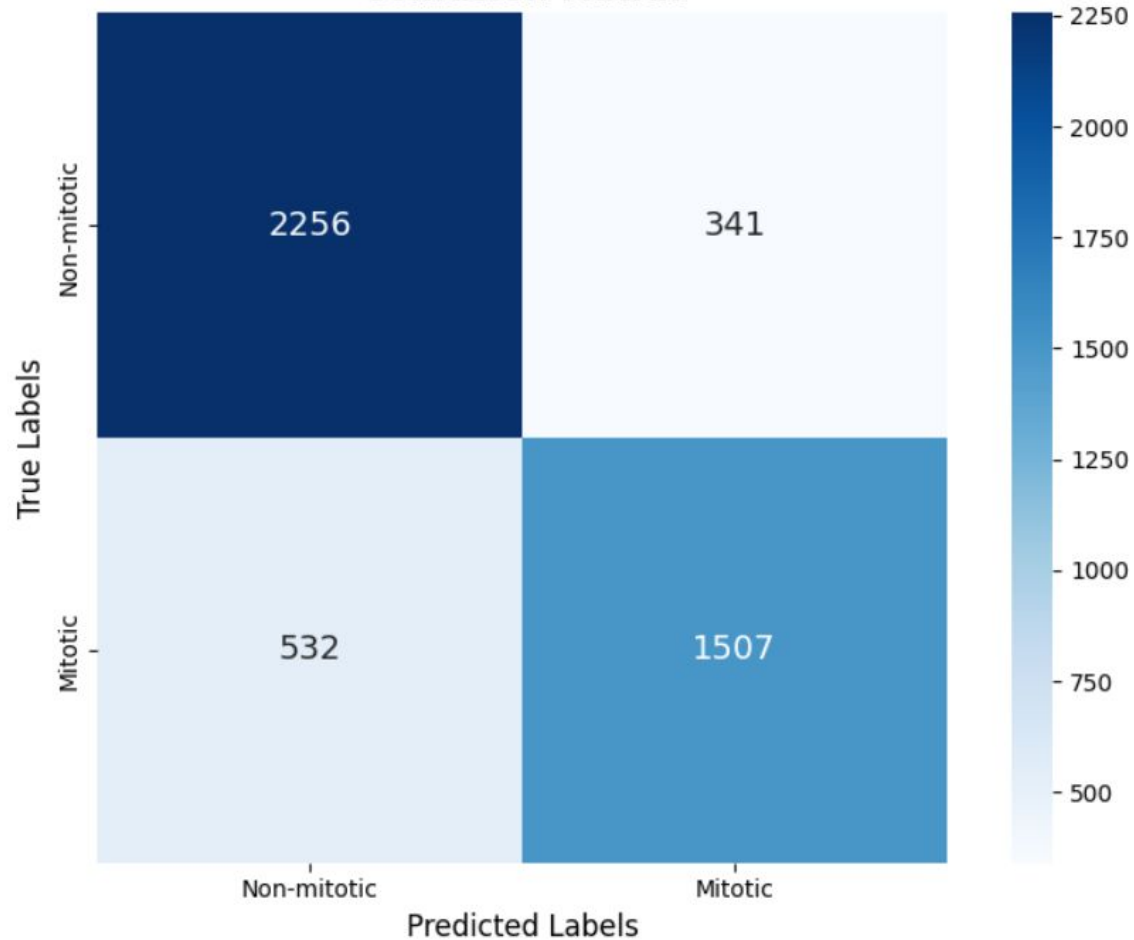
Goal:

- Feature extractor tries to fool the domain classifier
- Domain classifier tries to identify the slide source
- Layer penalizes features that reveal slide identity

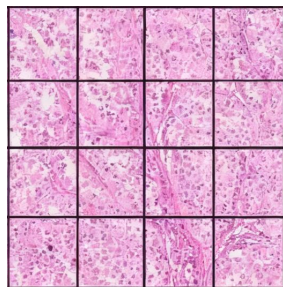
Control Results (No penalty)



Confusion Matrix



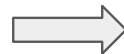
Multi-Domain Domain Adversarial Neural Network (DANN)



224x224x3 Patch
Extraction around
50x50 annotated
bbox



ResNet50 pretrained on
ImageNet, stripped of its final
classification layer



2048x1 Feature
Vector



(same features go to all 5 heads)

Mitosis Classifier -> Mitotic / Non-mitotic (TRAINED NORMALLY)
GRL -> Tumor Classifier -> 7 tumor types (ADVERSARIAL)
GRL -> Species Classifier -> Canine / Human (ADVERSARIAL)
GRL -> Origin Classifier -> 4 labs (ADVERSARIAL)
GRL -> Scanner Classifier -> 4 scanners (ADVERSARIAL)

Classifiers Head Structure

Mitotic head

2048-d feature vector

- > Flatten
- > Linear(2048 $\rightarrow$ 512)
- > ReLU
- > Dropout(0.5)
- > Linear(512 $\rightarrow$ 2)
- > Softmax $\rightarrow$ [Non-mitotic, Mitotic]

Domain Classifier Heads (4 same adversarial heads)

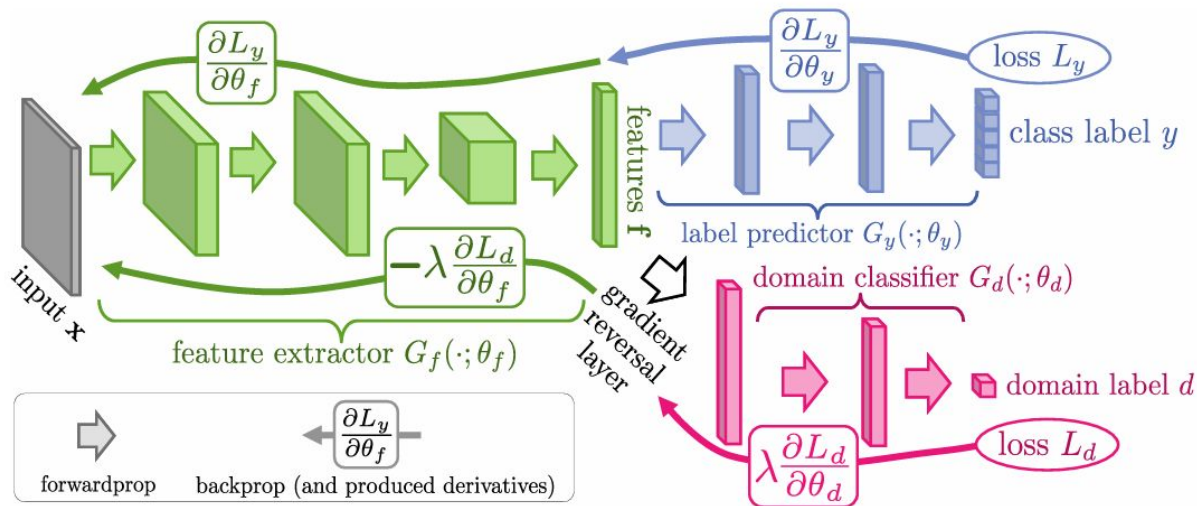
2048-d feature vector

- > Gradient Reversal Layer
- > Flatten
- > Linear(2048 $\rightarrow$ 512)
- > ReLU
- > Dropout(0.5)
- > Linear(512 $\rightarrow$ 2)
- > Softmax $\rightarrow$ [Non-mitotic, Mitotic]

Note: Each head has its own independent GRL so the adversarial pressure for each domain attribute is applied separately.

Our DANN architecture was mostly taken from: Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., & Lempitsky, V.S. (2015). Domain-Adversarial Training of Neural Networks. Journal of machine learning research.

Diagram of GRL



Following paper: $\lambda_p = \frac{2}{1 + \exp(-\gamma \cdot p)} - 1$, γ was set to 10 and $p = (\text{epoch} / \text{tot_epoch})$

"The domain adaptation parameter λ is initiated at 0 and is gradually changed to 1 using the following schedule... This strategy allows the domain classifier to be less sensitive to noisy signal at the early stages of the training procedure."

The Loss Function

Three components:

1. Mitosis loss (weighted cross-entropy):

- a. $L_{\text{mitosis}} = \text{CrossEntropyLoss}(\text{weight}=[1.0, 2.0])$
- b. The 2x weight on the mitotic class penalizes missed mitoses twice as heavily as false positives. This directly boosts recall.

2. Domain loss (unweighted cross-entropy):

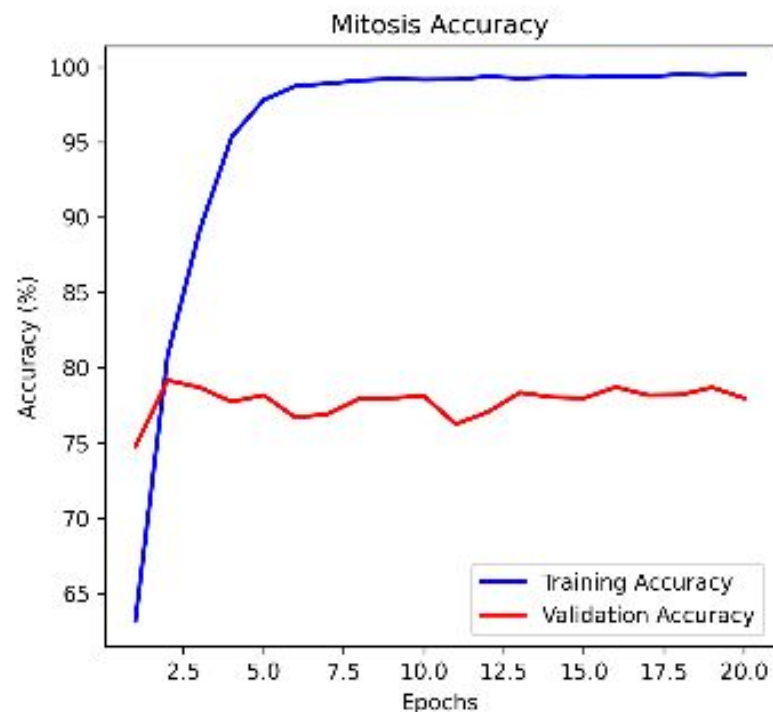
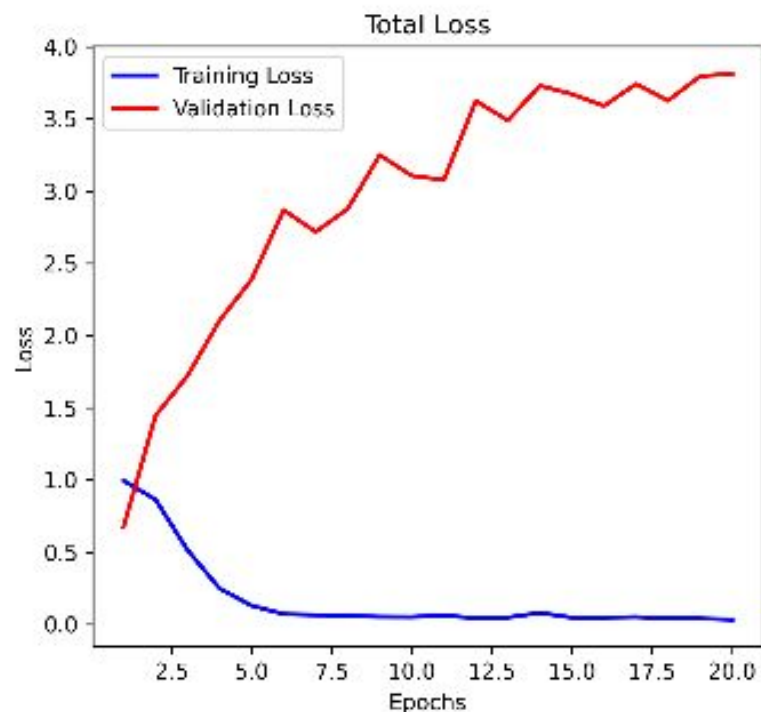
- a. $L_{\text{domains}} = (L_{\text{tumor}} + L_{\text{species}} + L_{\text{origin}} + L_{\text{scanner}}) / 4$

3. Adaptive weight

- a. $\text{running_mitosis} = 0.99 \times \text{running_mitosis} + 0.01 \times L_{\text{mitosis}}$
- b. $\text{running_domain} = 0.99 \times \text{running_domain} + 0.01 \times L_{\text{domains}}$
- c. $\text{adaptive_weight} = \text{running_mitosis} / \text{running_domain}$

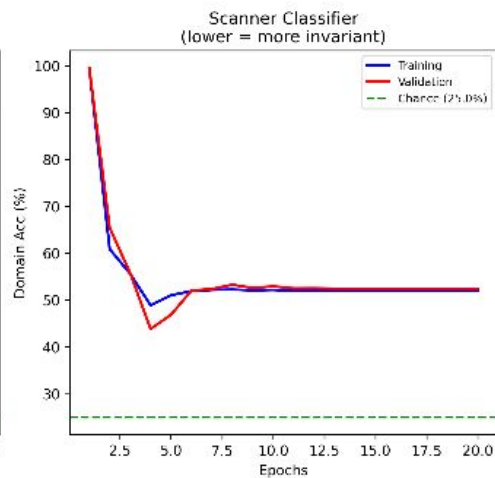
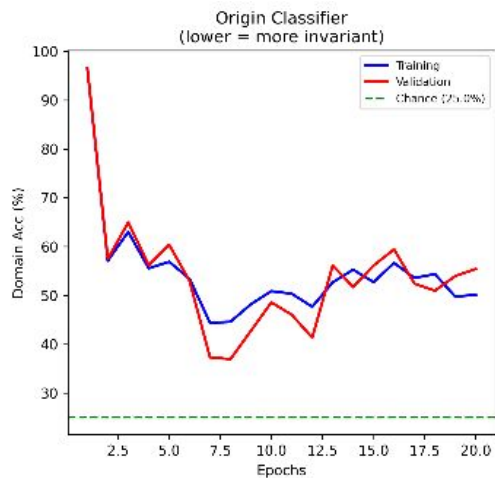
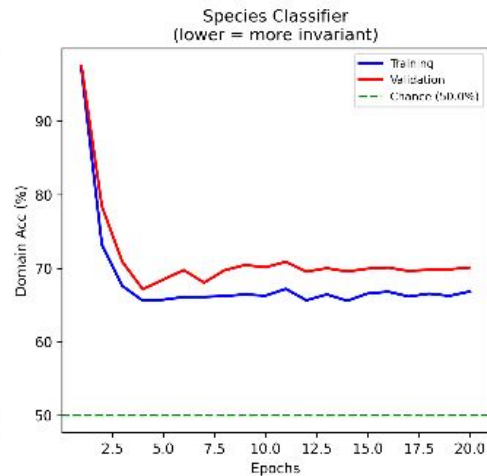
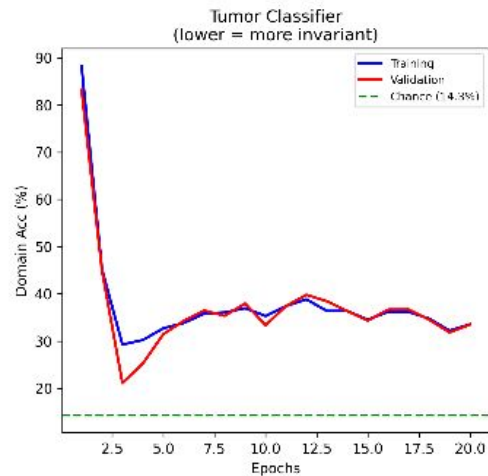
Final loss: $L_{\text{total}} = L_{\text{mitosis}} + \text{adaptive_weight} \times L_{\text{domains}}$

Results

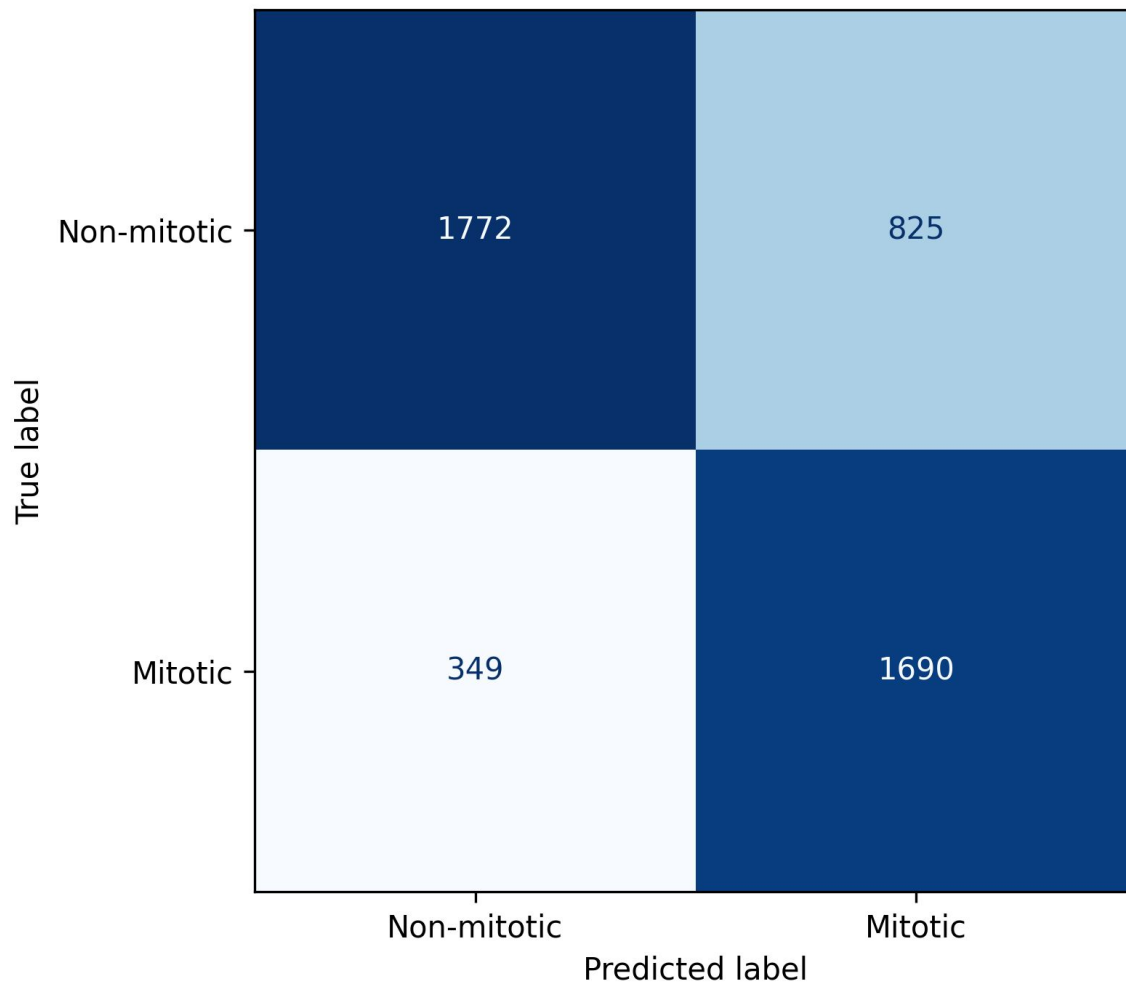


We are overfitting!

Domain Accuracy



Confusion Matrix - Best DANN Model



Comparison

	Control	DANN
Mitotic F1	0.78	0.74
Non-Mitotic F1	0.84	0.75
Mitotic Recall	0.74	0.83
Mitotic Precision	0.82	0.67

Clearly, mitotic precision suffers!

Image Augmentation

Biological interpretation: Mitotic figures vary in size across species and tissue types.

- different tumor types have different nuclear sizes

Stretching and shrinking directly attacks the morphological features the species and origin classifiers are latching onto!

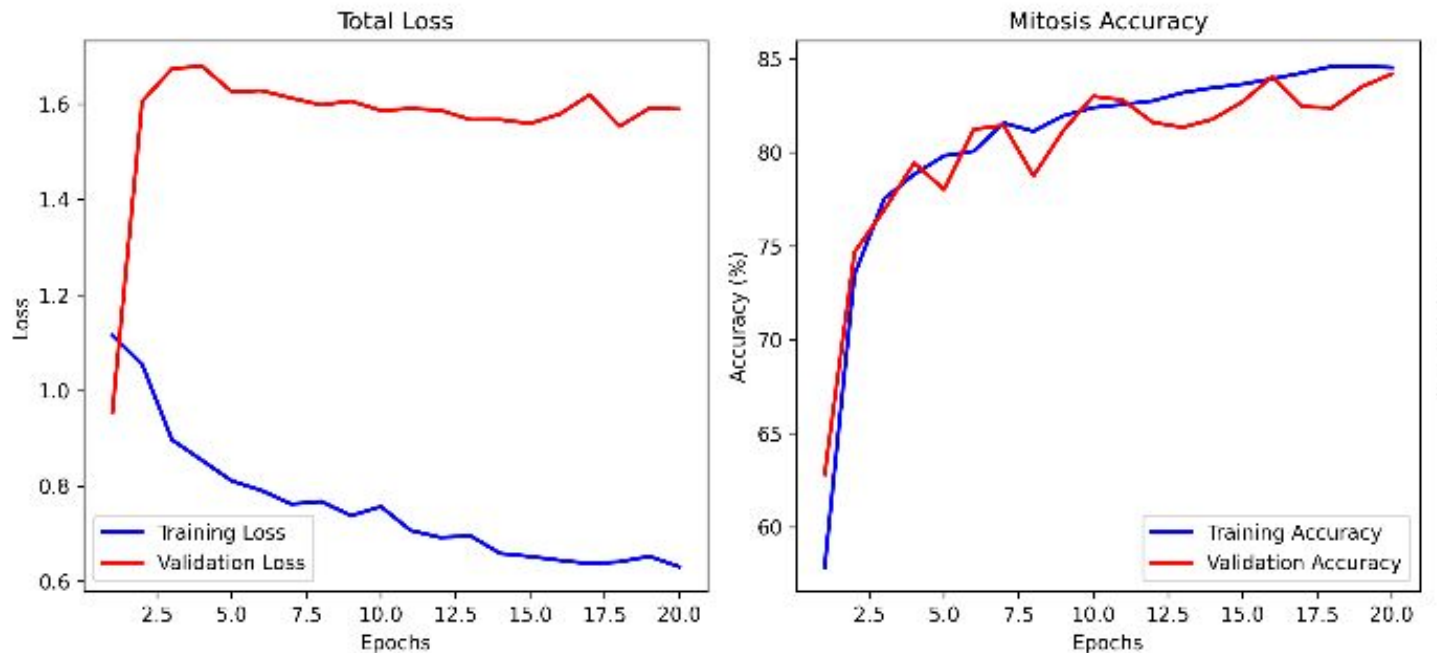
RandomResizedCrop: By randomly zooming in and out, the model sees mitotic figures at many different apparent sizes within training. This helps prevent species discrimination and resolution (some had 0.23 um/px while others had 0.25 um/px)

ElasticTransform: Deforms the image locally, simulating how cells stretch and compress differently depending on tissue preparation and fixation protocol across lab origins

RandomRotation(90) helps because mitotic figures have no canonical orientation. A cell in anaphase looks the same regardless of which way it's pointing, but the model shouldn't rely on orientation as a cue.

Note: No color augmentation because we were planning to use color deconvolution later.

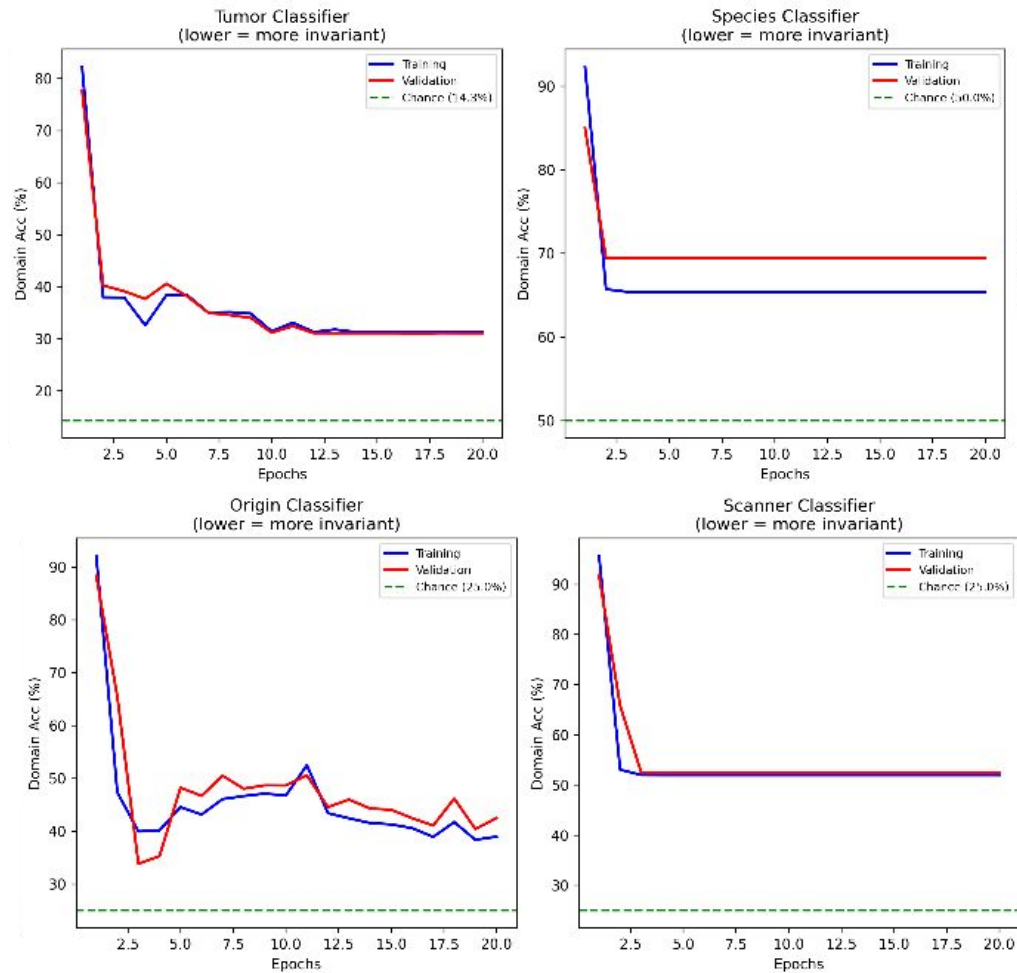
Results



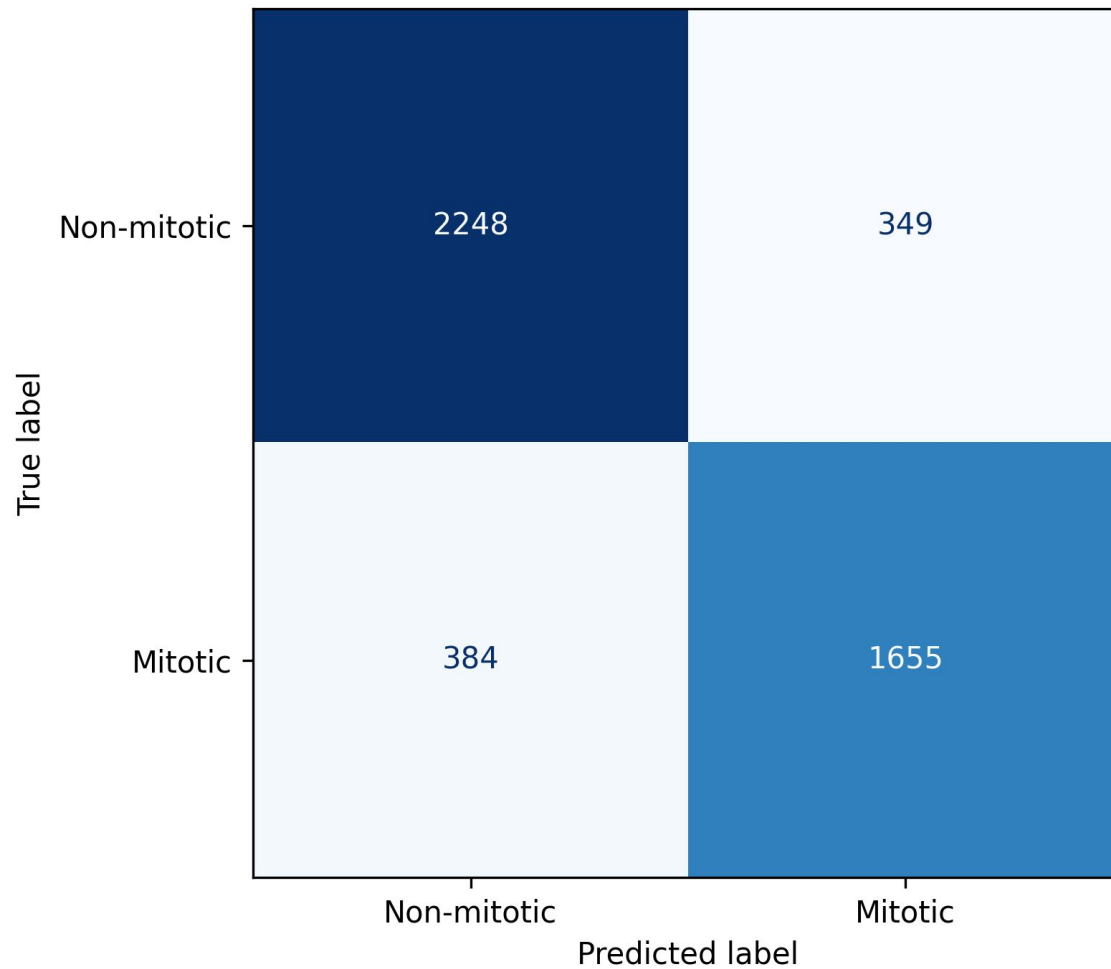
Note: Still hasn't stabilized at 20 epoch (took 8 hours to run).

Reduced overfitting!!!

Domain Accuracy



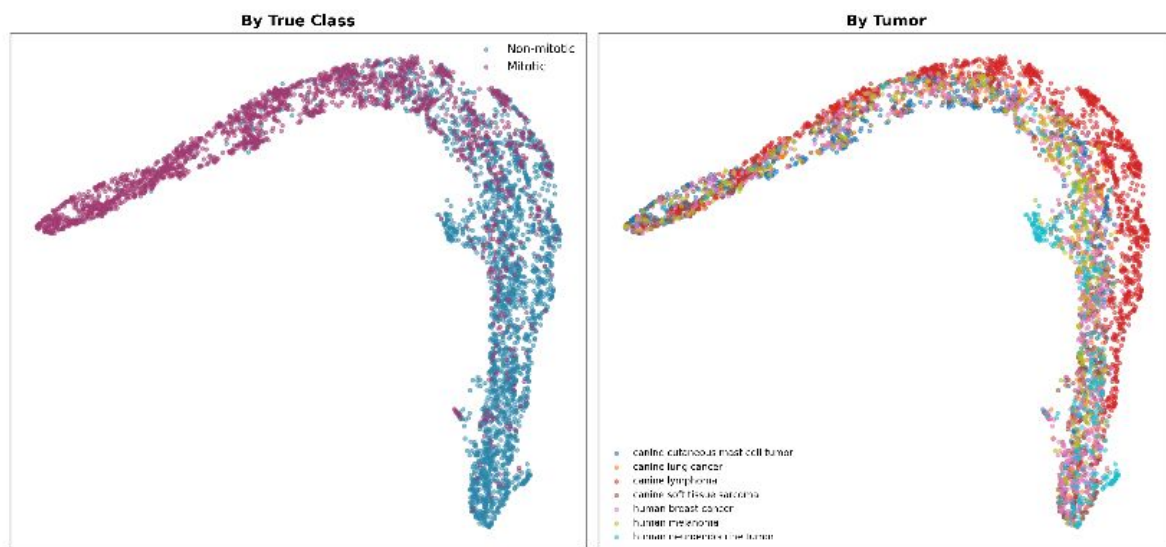
Confusion Matrix - Best DANN Model



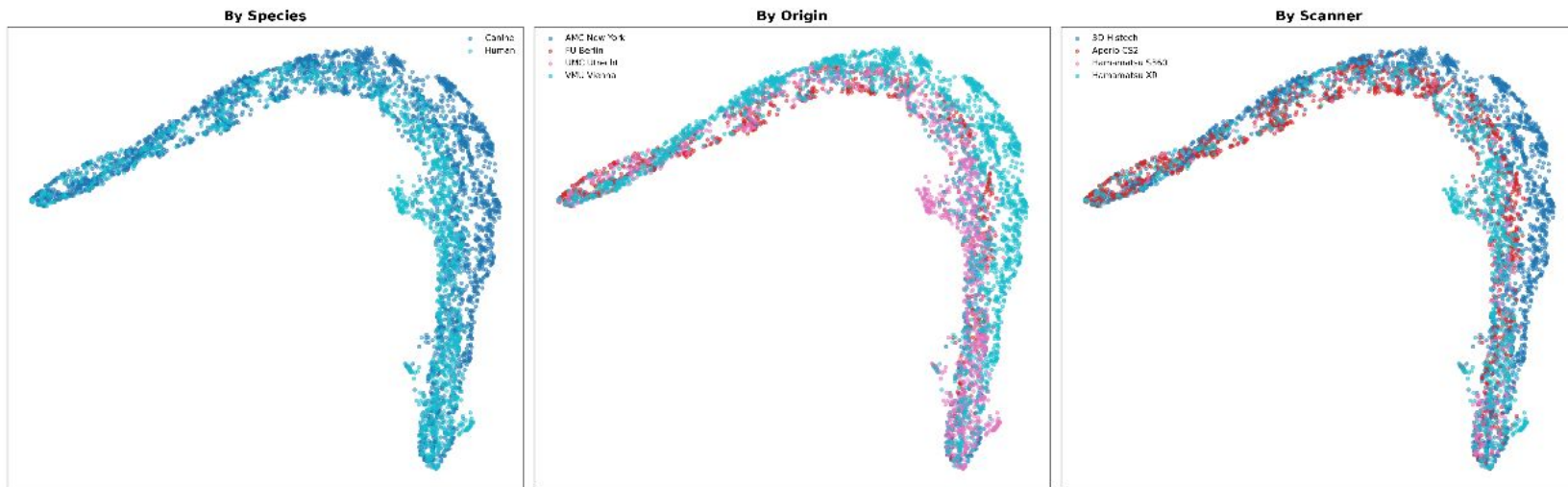
Comparison

	Control	DANN	DANN Image Aug
Mitotic F1	0.78	0.74	0.81
Non-Mitotic F1	0.84	0.75	0.87
Mitotic Recall	0.74	0.83	0.81
Mitotic Precision	0.82	0.67	0.83

UMAP of Multi-Domain DANN Feature Extractor (Best Model)



Better
intermixing



Next Steps:

- Incorporate color deconvolution into current pipeline
- Add confidence layer for ambiguous cases
- Adjust lambda to put more pressure

Thank you!